

Haolin Fan
CURRICULUM VITA

PERSONAL DATA

Born:	Oct 06, 2003	School:	Tsinghua University
Citizenship:	Chinese	Degree:	Mathematical Physics B.S.
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EDUCATION

B.S.	Tsinghua University	2022-2026
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Academic Interest

Focusing on constructing quantum field theory and string theory mathematically and using them to fix mathematical problems.

Recently, in collaboration with Professor Si Li, we discovered the D-module structure on $N=1$ superspace and classified the corresponding SUSY-invariant local functionals. Additionally, we constructed the BV complex of the Super-Yang-Mills theory and quantized it in the sense of homotopy renormalization. The detailed notes on these developments are now available on my GitHub.

Recently, I have also been focusing on quantum gravity like black hole and AdS/CFT.

PUBLICATIONS

works(details in my github <https://github.com/Roilkl/Note>)

1. "Quantization of $N = 1$ SYM theory"
2. "D-module in mirror symmetry"
3. "Short Note on Seiberg–Witten invariants"
4. "KZ equation and correspondence in Chern-Simons Theory"

OUTSTANDING LESSONS

Quantum Field Theory (2)	A
Topology	A
Complex Geometry	A-
String Theory	A
Topics in Derived Categories	A

Hobby

Tennis;Baseball;Ice hockey and Piano!

Ability

Have a good knowledge of basic Algebraic Geometry, Complex Geometry, Riemannian Geometry, Representation Theory, Functional Analysis, Algebraic Quantum Field Theory, BV formalism, SUSY and GR. Learning string theory, SUGRA, and the Derived Category this semester.